Comparative Research on GPU, its Operating Systems and their Application in AWS Cloud Platforms

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Module Code: CN7029

Module Title: Computer Architecture and Networks

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1. Introduction

Graphics Processing Units (GPUs) have expanded beyond dedicated graphics cards designed to accelerate graphics processing to become the key to high-performance computing, artificial intelligence, machine learning, or cloud-based applications. In contrast to the traditional Central Processing Unit (CPU), GPUs offer enormous processing power in the form of massive parallel or multi-user processing capabilities, such that they enable the processing of multiple calculations simultaneously in an efficient manner. Their use ensures that it is especially appropriate in image processing, deep learning, and scientific computations, as well as blockchain calculations.

Modern operating system (OS) and cloud computing providers like AWS and Azure have integrated GPUs to enable the virtualization of the GPUs and offer scalable and on-demand compute power. The proposed research is aimed at studying Linux GPU operating system commands, monitoring, and diagnostics, and real-life implementation of the GPU-supported computing in a cloud infrastructure.

Each coursework consists of four major tasks:

- Task 1: Linux GPU Operating System instructions to monitor and diagnose.
- Task 2. The practical implementation of cloud computing in AWS EC2 instances.
- Task 3: Reflective analysis of the GPU OS commands within cloud computing.
- Task 4: Conclusion, which summarizes the work and its main findings.

This report gives a comprehensive review about the chosen GPU commands along with their outputs and the stepwise procedure of implementing an EC2 instance in AWS. This discussion addresses the complementary aspects of Linux commands and cloud computing environments in providing efficient solutions based on the use of the GPUs.

2. Task 1: Linux GPU Operating System Commands

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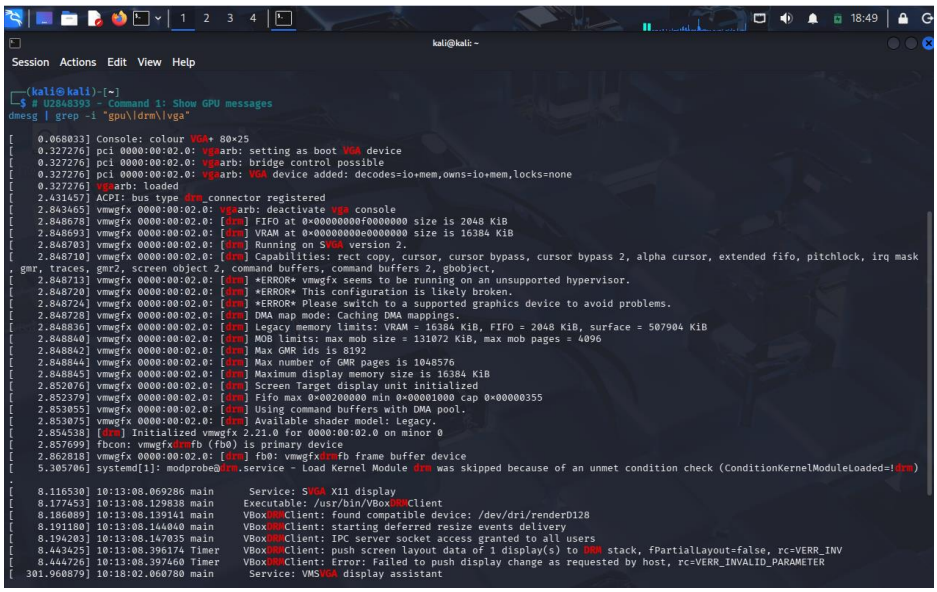

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┌──(kali㉿kali)-[~]
└─$ # U2848393 - Command 3: Show DRM GPU devices
ls /sys/class/drm/

card0            card0-Virtual-2  card0-Virtual-4  card0-Virtual-6  card0-Virtual-8  version
card0-Virtual-1  card0-Virtual-3  card0-Virtual-5  card0-Virtual-7  renderD128

┌──(kali㉿kali)-[~]
└─$
```

```
┌──(kali㉿kali)-[~]
└─$ # U2848393 - Command 4: Show GPU driver in use
lsmod | grep -i "drm\|vmwgfx\|vbox"

vboxsf                  49152  0
vboxguest               57344  7 vboxsf
vmwgfx                 491520  2
drm_ttm_helper          16384  2 vmwgfx
ttm                    135168  2 vmwgfx,drm_ttm_helper
drm_client_lib          12288  1 vmwgfx
drm_kms_helper         258048  3 vmwgfx,drm_ttm_helper,drm_client_lib
drm                    831488  8 vmwgfx,drm_kms_helper,drm_ttm_helper,drm_client_lib,ttm

┌──(kali㉿kali)-[~]
└─$
```

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┌──(kali㉿kali)-[~]
└─$ # U2848393 - Command 5: Show GPU device info via PCI
lspci -v | grep -A 10 -i "vga\|3d\|2d"

00:02.0 VGA compatible controller: VMware SVGA II Adapter (prog-if 00 [VGA controller])
        Subsystem: VMware SVGA II Adapter
        Flags: bus master, fast devsel, latency 64, IRQ 18
        I/O ports at d010 [size=16]
        Memory at e0000000 (32-bit, prefetchable) [size=16M]
        Memory at f0000000 (32-bit, non-prefetchable) [size=2M]
        Expansion ROM at 000c0000 [virtual] [disabled] [size=128K]
        Kernel driver in use: vmwgfx
        Kernel modules: vmwgfx

00:03.0 Ethernet controller: Intel Corporation 82540EM Gigabit Ethernet Controller (rev 02)
        Subsystem: Intel Corporation PRO/1000 MT Desktop Adapter

┌──(kali㉿kali)-[~]
└─$
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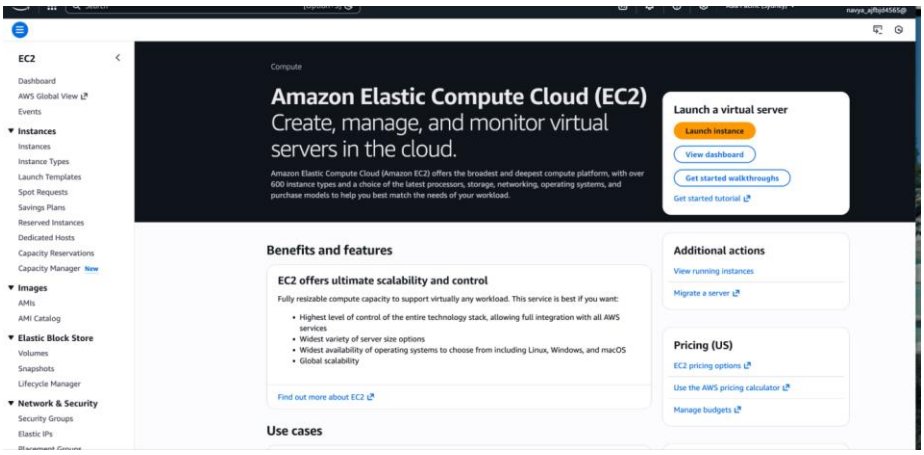

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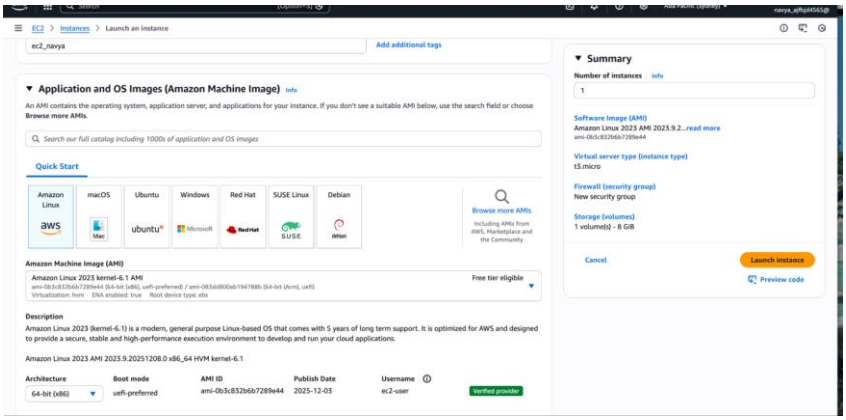

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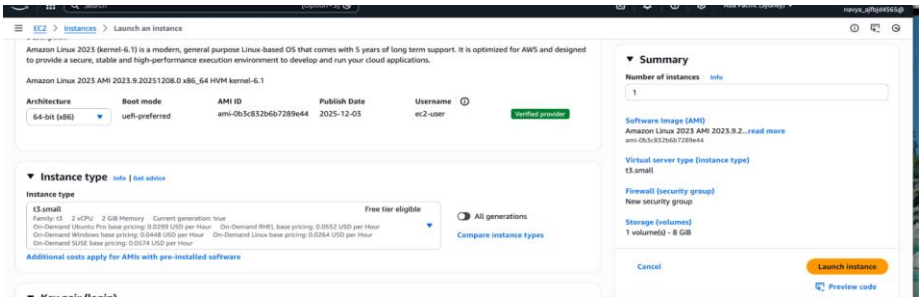

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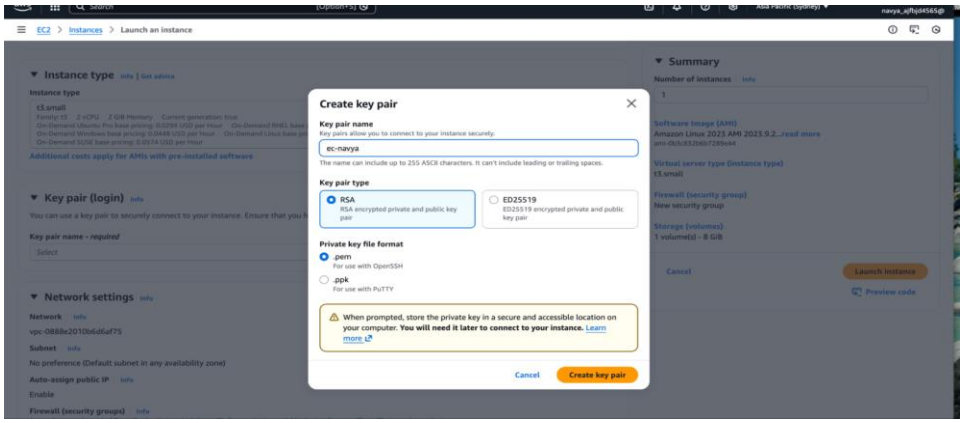

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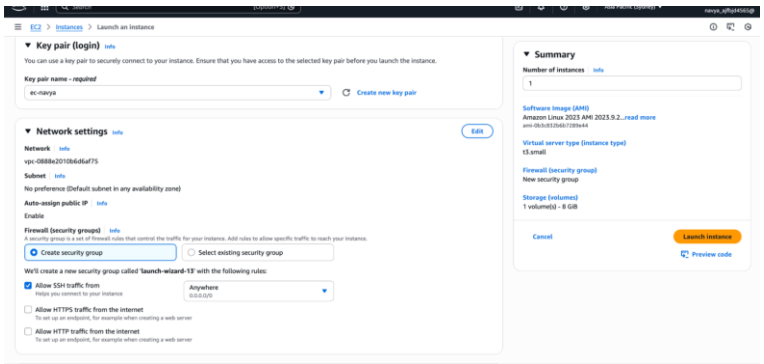

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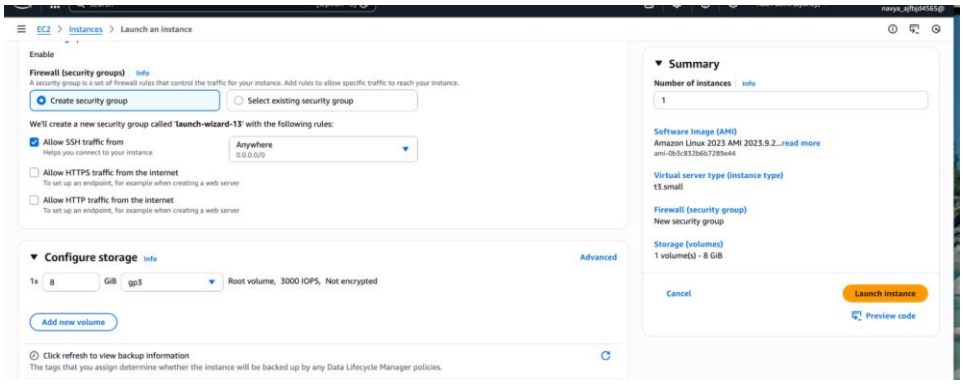

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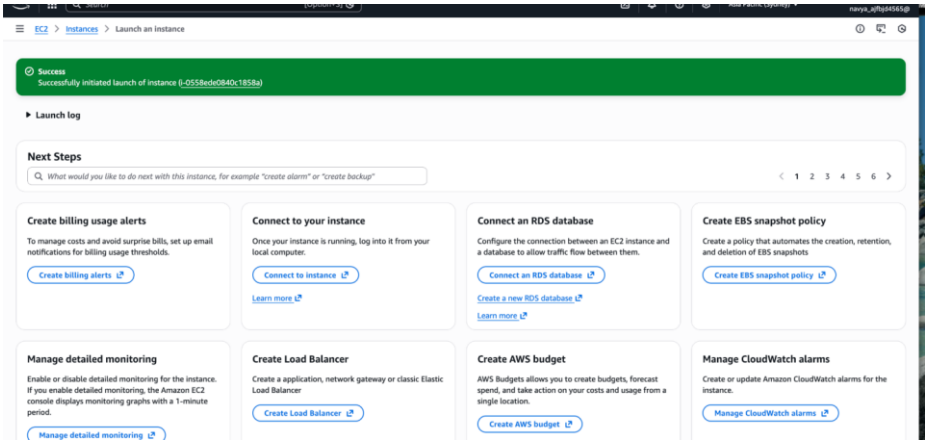

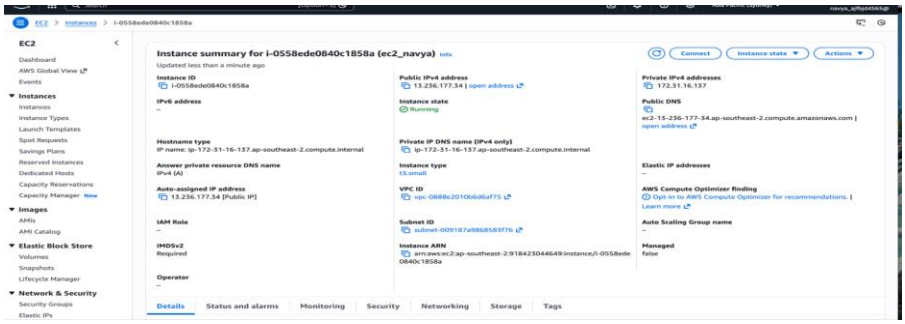

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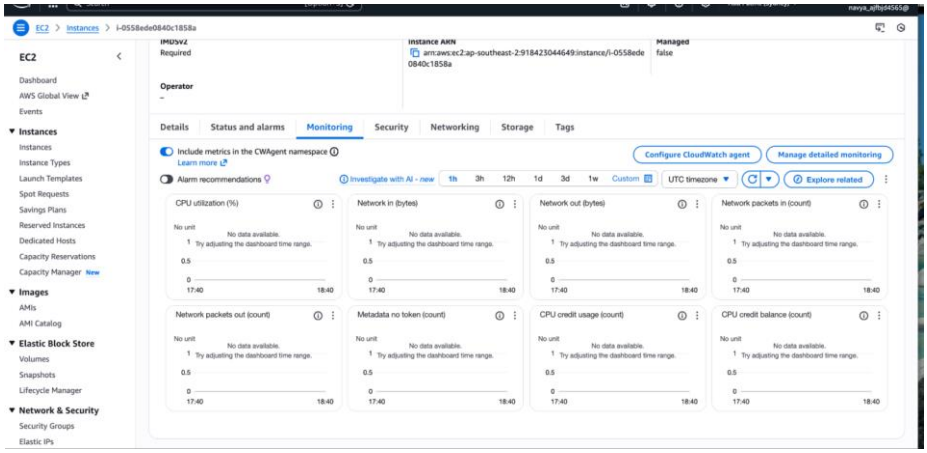

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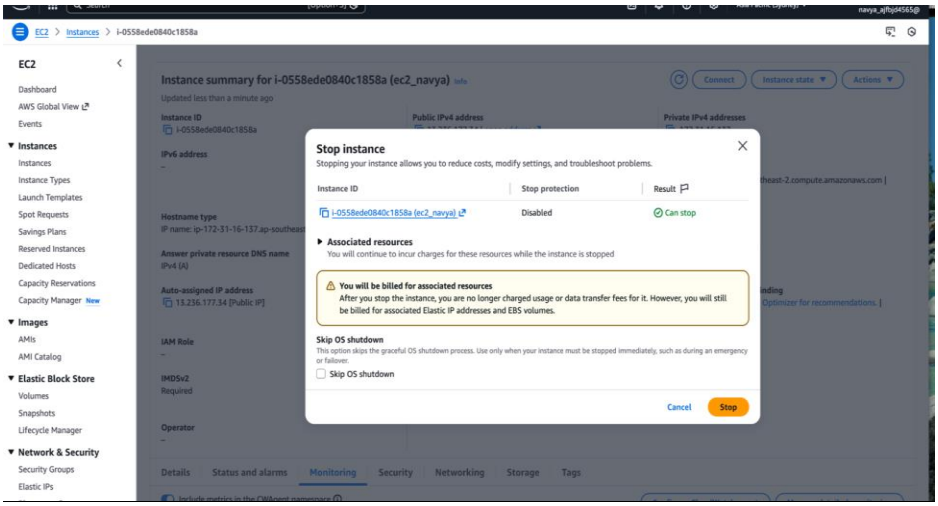

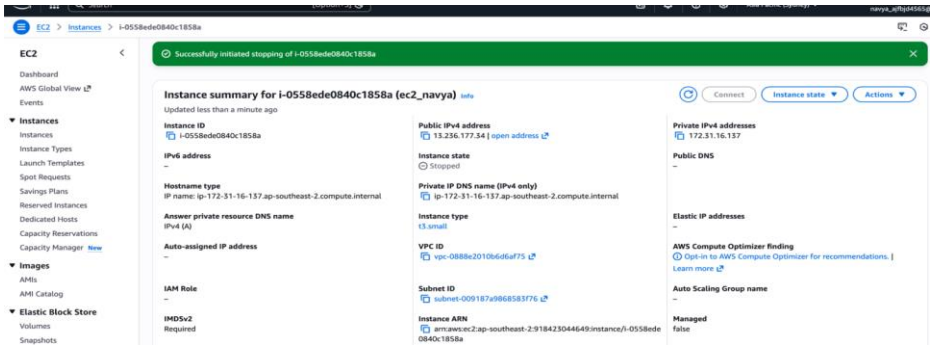

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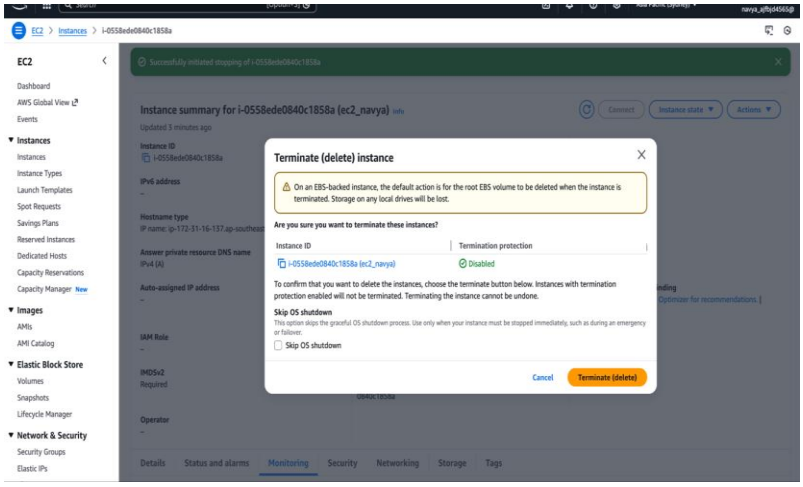

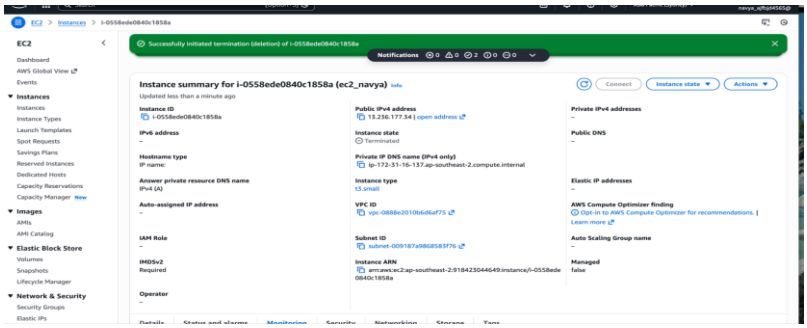

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